Chemical kinetics

The branch of Physical chemistry which deals with the rate of reactions is called Chemical Kinetics.

The study of Chemical Kinetics includes:
(1) The rate of the reactions and rate laws.
(2) The factors as temperature, pressure, concentration and catalyst, which influences the rate of a reaction.
(3) The mechanism or the sequence of steps by which a reaction occurs.

REACTION RATE

The rate of reactions is defined as the change in concentration of any of reactant or products per unit time.

$$A \longrightarrow B$$

For the given reaction the rate of reaction may be equal to the rate of disappearance of A which is equal to the rate of appearance of B. Thus rate of reaction = rate of disappearance of *A*= rate of appearance of *B*

or $$\text{rate} = -\frac{d[A]}{dt}$$

$$= +\frac{d[B]}{dt}$$

where [ ] represents the concentration in moles per litre whereas 'd' represents infinitesimally small change in concentration. Negative sign shows the concentration of the reactant A decreases whereas the positive sign indicates the increase in concentration of the product B.

UNITS OF RATE

Reactions rate has the units of concentration divided by time. We express concentrations in moles per litre (mol/litre or mol/1 or mol 1^{-1}) but time may be given in any convenient unit second (s), minutes (min), hours (h), days (d) or possible years. Therefore, the units of reaction rates may be

mole/litre sec	or	mol 1^{-1} s
mole/litre min	or	mol 1^{-1} min^{-1}
mole/litre hour	or	mol 1^{-1} h^{-1} and, so on

Instantaneous Rate of Reaction

Thus at any time the instantaneous rate is equal to the slope of a straight line drawn tangent to the curve at that time. For example, in Fig. 20.2 the instantaneous rate at 10 seconds is found to be 0.0022 mol l–1 s–1.

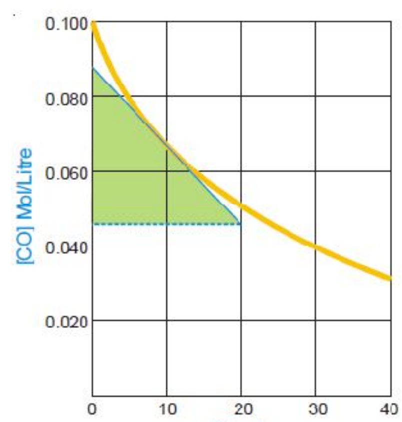


Figure 20.2
The slope of tangent at 10 seconds is equal to the instantaneous rate.

RATE LAWS

the rate of a reaction is directly proportional to the reactant concentrations, each concentration being raised to some power.

Thus for a substance A undergoing reaction,

rate $\quad$ rate $\propto [A]^n$

or $\quad$ rate $= k\,[A]^n$ $\quad$ ...(1)

For a reaction

$$2A + B \longrightarrow \text{products}$$

the reaction rate with respect to A or B is determined by varying the concentration of one reactant, keeping that of the other constant. Thus the rate of reaction may be expressed as

$$\text{rate} = k\,[A]^m\,[B]^n \qquad ...(2)$$

Expressions such as (1) and (2) tell the relation between the rate of a reaction and reactant concentrations.

An expression which shows how the reaction rate is related to concentrations is called the rate law or rate equation.

The power (exponent) of concentration n or m in the rate law is usually a small whole number integer (1, 2, 3) or fractional. The proportionality constant k is called the **rate constant** for the reaction.

Examples of rate law :

	REACTIONS	RATE LAW
(1)	$2N_2O_5 \longrightarrow 4NO_2 + O_2$	rate $= k\,[N_2O_5]$
(2)	$H_2 + I_2 \longrightarrow 2HI$	rate $= k\,[H_2]\,[I_2]$
(3)	$2NO_2 \longrightarrow 2NO + O_2$	rate $= k\,[NO_2]^2$
(4)	$2NO + 2H_2 \longrightarrow N_2 + 2H_2O$	rate $= k\,[H_2]\,[NO]^2$

It is apparent that the rate law for a reaction must be determined by experiment. It cannot be written by merely looking at the equation with a background of our knowledge of Law of Mass Action. However, for some elementary reactions the powers in the rate law may correspond to coefficients in the chemical equation. **But usually the powers of concentration in the rate law are different from coefficients.** Thus for the reaction (4) above, the rate is found to be proportional to $[H_2]$ although the quotient of H_2 in the equation is 2. For NO the rate is proportional to [NO]2 and power '2' corresponds to the coefficient.

ORDER OF A REACTION

The order of a reaction is defined as the sum of the powers of concentrations in the rate law.

Let us consider the example of a reaction which has the rate law

rate $= k\,[A]^m\,[B]^n$ ...(1)

The order of such a reaction is $(m + n)$. The order of a reaction can also be defined with respect to a single reactant. Thus the reaction order with respect to A is m and with respect to B it is n. The **overall order of reaction** $(m + n)$ may range from 1 to 3 and can be fractional.

Examples of reaction order :

RATE LAW	REACTION ORDER
rate = $k\,[N_2O_5]$	1
rate = $k\,[H_2]\,[I_2]$	$1 + 1 = 2$
rate = $k\,[NO_2]^2$	2
rate = $k\,[[H_2]\,[NO]^2$	$1 + 2 = 3$
rate = $k\,[CHCl_3]\,[Cl_2]^{1/2}$	$1 + \frac{1}{2} = 1\frac{1}{2}$

Reactions may be classified according to the order. If in the rate law (1) above

$m + n = 1$, it is **first order reaction**

$m + n = 2$, it is **second order reaction**

$m + n = 3$, it is **third order reaction**

ZERO ORDER REACTION

A zero order reaction is one whose rate is independent of concentration.
For example, the rate law for the reaction

$NO_2 + CO \longrightarrow NO + CO_2$ at 200° C is rate = $k\,[NO_2]^2$

Here the rate does not depend on [CO], so this is not included in the rate law and the power of [CO] is understood to be zero. The reaction is **zeroth order** with respect to CO. The reaction is second order with respect to $[NO_2]$. The overall reaction order is 2 + 0 = 2.

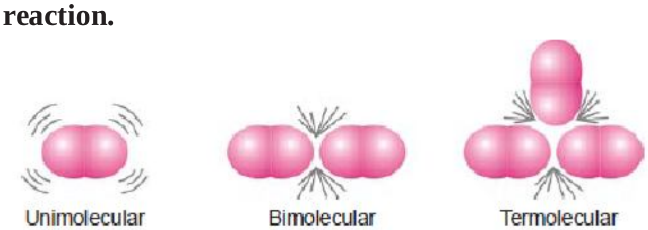
MOLECULARITY OF A REACTION

The total number of molecules or atoms which take part in a reaction as represented by the chemical equation is known as the **molecularity of reaction.**

Chemical reactions may be classed into two types:
(*a*) Elementary reactions
(*b*) Complex reactions
An **elementary reaction** is a simple reaction which occurs in a single step.
A **complex reaction** is that which occurs in two or more steps.

Molecularity of an Elementary Reaction

The molecularity of an elementary reaction is defined as : **the number of reactant molecules involved in a reaction.**

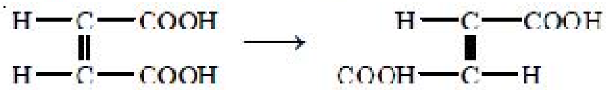


(*a*) **Unimolecular reactions** : (molecularity = 1)

$$A \longrightarrow \text{product}$$

Examples are : (*i*) $$Br_2 \longrightarrow 2Br$$

(*ii*)



maleic acid fumaric acid

(*b*) **Bimolecular reactions** : (molecularity = 2)

$$A + B \longrightarrow \text{products}$$
$$A + A \longrightarrow \text{products}$$

Examples are :

(*i*) $$\underset{\text{Ethyl acetate}}{CH_3COOC_2H_5} + H_2O \longrightarrow \underset{\text{acetic acid}}{CH_3COOH} + \underset{\text{ethyl alcohol}}{C_2H_5OH}$$

(*ii*) $$2HI \longrightarrow H_2 + I_2$$

(*c*) **Termolecular reactions** . (molecularity = 3)

$$A + B + C \longrightarrow \text{products}$$

Examples are :

$$2NO + O_2 \longrightarrow 2NO_2$$
$$2NO + Cl_2 \longrightarrow 2NOCl$$

MOLECULARITY *VERSUS* ORDER OF REACTION

The total number of molecules or atoms which take part in a reaction as represented by the chemical equation is known as the **molecularity of reaction.**

The sum of the powers to which the concentrations are raised in the rate law is known as the **order of reaction.**

Molecularity and Order are Identical for Elementary Reactions or Steps

The rate of an elementary reaction is proportional to the number of collisions between molecules (or atoms) of reactions. The number of collisions in turn is proportional to the concentration of each reactant molecule (or atom). Thus for a reaction.

$$2A + B \longrightarrow \text{products}$$
$$\text{rate} \propto [A]\,[A]\,[B]$$

or $$\text{rate} = k\,[A]^2\,[B] \qquad (\textit{rate law})$$

Two molecules of A and one molecule of B are participating in the reaction and, therefore, molecularity of the reaction is $2 + 1 = 3$. The sum of powers in the rate law is $2 + 1$ and hence the reaction order is also 3. Thus **the molecularity and order for an elementary reaction are equal.**

TABLE 20.1. MOLECULARITY AND ORDER FOR ELEMENTARY REACTIONS.

Reactions	Molecularity	Rate law	Order
$A \longrightarrow$ products	1	rate = k [A]	1
$A + A \longrightarrow$ products	2	rate = $k\,[A]^2$	2
$A + B \longrightarrow$ products	2	rate = k [A] [B]	2
$A + 2B \longrightarrow$ products	3	rate = $k[A]\,[B]^2$	3
$A + B + C \longrightarrow$ products	3	rate = k [A] [B] [C]	3

Differences Between Order and Molecularity

Order of a Reaction	Molecularity of a Reaction
1. It is the sum of powers of the concentration terms in the rate law expression	1. It is number of reacting species undergoing simultaneous collision in the elementary or simple reaction
2. It is an **experimentally** determined value.	2. It is a **theoretical** concept.
3. It can have **fractional** value.	3. It is always a **whole number.**
4. It can assume **zero value.**	4. **It can not** have **zero value.**
5. Order of a reaction **can change** with the conditions such as pressure, temperature, concentration	5. Molecularity is **invariant** for a chemical equation.

Collision Theory and Reaction Rate Expression

Taking into account the two postulates of the collision theory, the reaction rate for the elementary process.

$$A+B \longrightarrow C+D$$

is given by the expression

$$\text{rate} = f \times p \times z$$

where f = fraction of molecules which possess sufficient energy to react; p = probable fraction of collisions with effective orientations, and z – collision frequency.

COLLISION THEORY OF REACTION RATES

According to this theory, **a chemical reaction takes place only by collisions between the reacting molecules.** But not all collisions are effective. Only a small fraction of the collisions produce a reaction. The two main conditions for a collision between the reacting molecules to be productive are :

(1) The colliding molecules must posses sufficient kinetic energy to cause a reaction.

(2) The reacting molecules must collide with proper orientation.

Now let us have a closer look at these two postulates of the collision theory.

(1) The molecules must collide with sufficient kinetic energy

Let us consider a reaction

$$A-A \;+\; B-B \longrightarrow 2A-B$$

A chemical reaction occurs by breaking bonds between the atoms of the reacting molecules and forming new bonds in the product molecules. The energy for the breaking of bonds comes from the kinetic energy possessed by the reacting molecules before the collision. Fig. 20.9 shows the energy of molecules A_2 and B_2 as the reaction $A_2 + B_2 \rightarrow 2AB$ progresses.

Temperature Dependence of Reaction Rate and Arrhenius Equation

We know that the kinetic energy of a gas is directly proportional to its temperature. Thus as the temperature of a system is increased, more and more molecules will acquire necessary energy greater that E_a to cause productive collisions. This increases the rate of the reaction.

In 1889, Arrhenius suggested as simple relationship between the rate constant, k, for a reaction and the temperature of the system.

$$k = A\, e^{-E_a/RT} \qquad (1)$$

This is called the **Arrhenius equation** in which A is an experimentally determined quantity, E_a is

the activation energy, R is the gas constant, and T is Kelvin temperature.

Taking natural logs of each side of the Arrhenius equation, it can be put in a more useful form :

$$\ln k = -\frac{E_a}{RT} + \ln A \qquad ...(2)$$

$$\log k = \frac{-E_a}{2.303\,RT} + \log A \qquad ...(3)$$

If k_1 and k_2 are the values of rate constants at temperatures T_1 and T_2 respectively, we can derive

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303\,R}\left[\frac{T_2 - T_1}{T_1 T_2}\right] \qquad ...(4)$$

Arrhenius equation is valuable because it can be used to calculate the activation energy, E_a if the experimental value of the rate constant, k, is known.

Transition state theory

reactant molecules does not really causes a reaction. During the collision, **the reactant molecules form a transition state or activated complex which decomposes to give the products.**

Thus,

$$\underbrace{A + B{-}C}_{\text{Reactants}} \longrightarrow \underbrace{ABC^{\ddagger}}_{\text{Activated complex}} \longrightarrow \underbrace{A{-}B + C}_{\text{Products}}$$

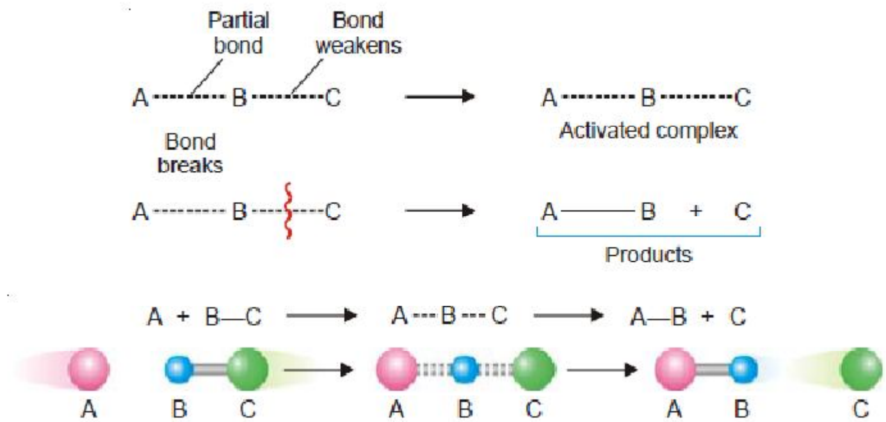
The double dagger superscript (‡) is used to identify the activated complex.

The transition state theory may be summarised as follows :

(1) In a collision, the fast approaching reactant molecules (A and BC) slow down due to gradual repulsion between their electron clouds. In the process **the kinetic energy of the two molecules is converted into potential energy.**

(2) As the molecules come close, **the interpenetration of their electron clouds occurs which allows the rearrangement of valence electrons.**

(3) A partial bond is formed between the atoms A and B with corresponding weakening of B – C bond. This leads to formation of an **activated complex or transition state.** The activated complex is momentary and decomposes to give the products (A–B + C)



The activated complex theory may be illustrated by the **reaction energy diagram** (Fig. 20.13).

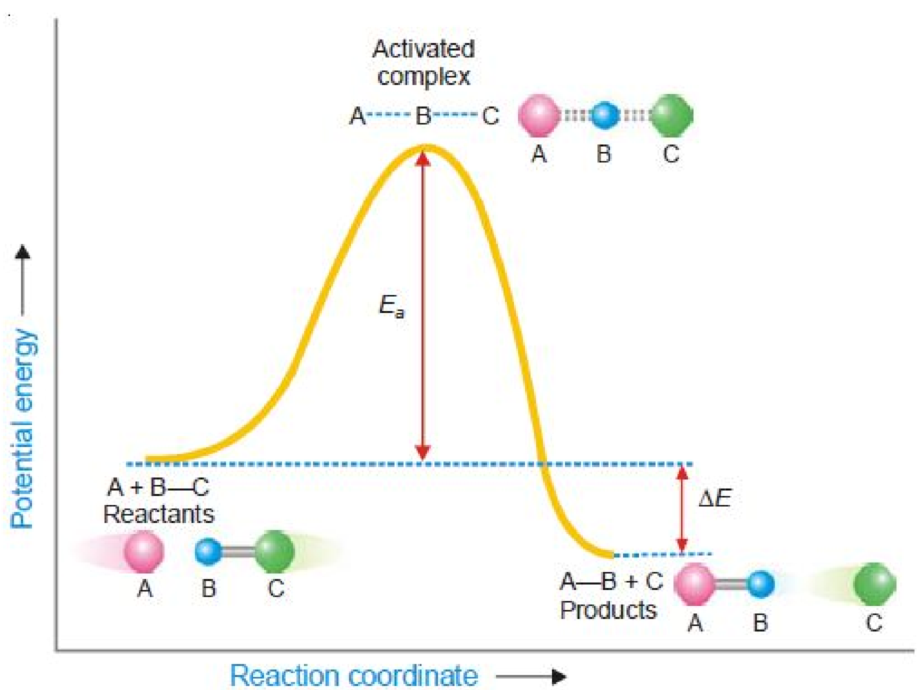


Figure 20.13

Change of potential energy during a collision between the reactant molecules for an exothermic reaction.

FIRST ORDER REACTIONS

Let us consider a first order reaction

$$A \longrightarrow \text{products}$$

Suppose that at the beginning of the reaction ($t = 0$), the concentration of A is a moles litre^{-1}. If after time t, x moles of A have changed, the concentration of A is $a - x$. We know that for a first order reaction, the rate of reaction, dx/dt, is directly proportional to the concentration of the reactant. Thus,

$$\frac{dx}{dt} = k(a - x)$$

or

$$\frac{dx}{a - x} = k\,dt \qquad \text{...(1)}$$

Integration of the expression (1) gives

$$\int \frac{dx}{a - x} = \int k\,dt$$

or

$$-\ln(a - x) = kt + \mathrm{I} \qquad \text{...(2)}$$

where I is the constant of integration. The constant k may be evaluated by putting $t = 0$ and $x = 0$.

Thus,

$$\mathrm{I} = -\ln a$$

Substituting for I in equation (2)

$$\ln\frac{a}{a - x} = kt \qquad \text{...(3)}$$

or

$$k = \frac{1}{t}\ln\frac{a}{a - x}$$

Changing into common logarithms

$$k = \frac{2.303}{t}\log\ \frac{a}{a - x} \qquad \text{...(4)}$$

The value of k can be found by substituting the values of a and $(a - x)$ determined experimentally at time interval t during the course of the reaction.

Sometimes the integrated rate law in the following form is also used :

$$k = \frac{2.303}{t_2 - t_1}\log\ \frac{(a - x_1)}{(a - x_2)}$$

where x_1 and x_2 are the amounts decomposed at time intervals t_1 and t_2 respectively from the start.

SECOND ORDER REACTIONS

Let us take a second order reaction of the type

$$2A \longrightarrow \text{products}$$

Suppose the initial concentration of A is a moles litre^{-1}. If after time t, x moles of A have reacted, the concentration of A is $(a - x)$. We know that for such a second order reaction, rate of reaction is proportional to the square of the concentration of the reactant. Thus,

$$\frac{dx}{dt} = k\,(a - x)^2 \qquad ...(1)$$

where k is the rate constant, Rearranging equation (1), we have

$$\frac{dx}{(a - x)^2} = k\,dt \qquad ...(2)$$

On integration, it gives

$$\frac{1}{a - x} = kt + I \qquad ...(3)$$

where I is is integration constant. I can be evaluated by putting $x = 0$ and $t = 0$. Thus,

$$I = \frac{1}{a} \qquad ...(4)$$

Substituting for I in equation (3)

$$\frac{1}{a - x} = kt + \frac{1}{a}$$

$$kt = \frac{1}{a - x} - \frac{1}{a}$$

Thus $$k = \frac{1}{t} \cdot \frac{x}{a(a - x)}$$

This is the integrated rate equation for a second order reaction.

THIRD ORDER REACTIONS

Let us consider a simple third order reaction of the type

$$3A \longrightarrow \text{products}$$

Let the initial concentration of A be a moles litre^{-1} and after time t, x, moles have reacted. Therefore, the concentration of A becomes $(a - x)$. The rate law may be written as :

$$\frac{dx}{dt} = k(a - x)^3 \quad ...(1)$$

Rearranging equation (1), we have

$$\frac{dx}{(a - x)^3} = k\,dt \quad ...(2)$$

On integration, it gives

$$\frac{1}{2(a - x)^2} = kt + I \quad ...(3)$$

where I is the integration constant. I can be evaluated by putting $x = 0$ and $t = 0$. Thus,

$$I = \frac{1}{2a^2}$$

By substituting the value of I in (3), we can write

$$kt = \frac{1}{2(a - x)^2} - \frac{1}{2a^2}$$

Therefore,

$$k = \frac{1}{t} \cdot \frac{x(2a - x)}{2a^2(a - x)^2}$$

This is the integrated rate equation for a third order reaction.

HALF–LIFE OF A REACTION

Reaction rates can also be expressed in terms of **half-life or half-life period.** It is defined as : **the time required for the concentration of a reactant to decrease to half its initial value.**

In other words, half-life is the time required for one-half of the reaction to be completed. It is represented by the symbol $t_{1/2}$ or $t_{0.5}$

Calculation of Half-life of a First order Reaction

The integrated rate equation (4) for a first order reaction can be stated as :

$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$$

where $[A]_0$ is initial concentration and [A] is concentration at any time t. Half-life, $t_{1/2}$, is time when initial concentration reduces to $\frac{1}{2}$ *i.e.*,

$$[A] = \frac{1}{2}[A]_0$$

Substituting values in the integrated rate equation, we have

$$k = \frac{2.303}{t_{1/2}} \log \frac{[A]_0}{1/2[A]_0} = \frac{2.303}{t_{1/2}} \log 2$$

or

$$t_{1/2} = \frac{2.303}{k} \log 2 = \frac{2.303}{k} \times 0.3010$$

or

$$t_{1/2} = \frac{0.693}{k}$$

It is clear from this relation that :

(1) half-life for a first order reaction is **independent of the initial concentration.**

(2) it is **inversely proportional ot k,** the rate-constant.

SOLVED PROBLEM 5. For a certain first order reaction $t_{0.5}$ is 100 sec. How long will it take for the reaction to be completed 75% ?

SOLUTION

Calculation of k

For a first order reaction

$$t_{1/2} = \frac{0.693}{k}$$

or

$$100 = \frac{0.693}{k}$$

$\therefore$

$$k = \frac{0.693}{100} = 0.00693 \text{ sec}^{-1}$$

Calculation of time for 75% completion of reaction

The integrated rate equation for a first order reaction is

$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$$

or

$$t = \frac{2.303}{k} \log \frac{[A]_0}{[A]}$$

When $\frac{3}{4}$ initial concentration has reacted, it is reduced to $\frac{1}{4}$

Substituting values in the rate equation

$$t_{3/4} = \frac{2.303}{0.00693} \log \frac{[A]_0}{\frac{1}{4}[A]_0}$$

$$= \frac{2.303}{0.00693} \log 4 = \mathbf{200\ sec}$$

SOLVED PROBLEM 1. Compound A decomposes to form B and C the reaction is first order. At 25°C the rate constant for the reaction is 0.450 s^{-1}. What is the half-life of A at 25°C ?

SOLUTION

We know that for a first order reaction, half life $t_{1/2}$, is given by the expression

$$t_{1/2} = \frac{0.693}{k}$$

where k = rate constant

Substituting the value of $k = 0.450\ s^{-1}$, we have

$$t_{1/2} = \frac{0.693}{0.450\ s^{-1}} = \mathbf{1.54\ s}$$

Thus half-life of the reaction $A \rightarrow B + C$ is 1.54 seconds.